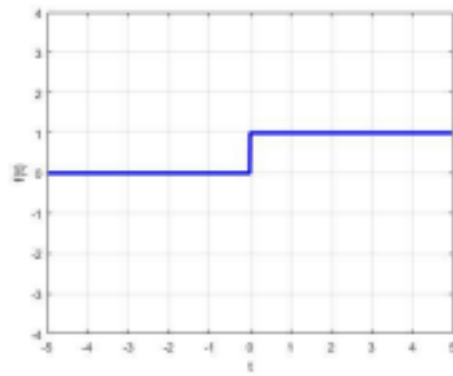
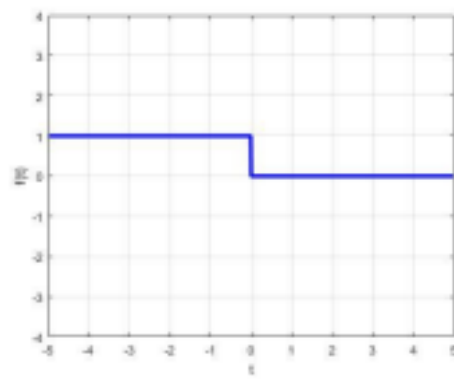


1 Sinais

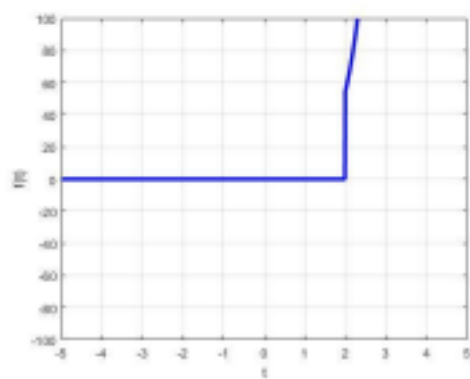
1)



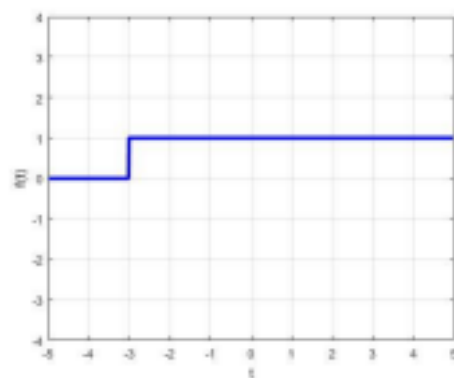
(a)



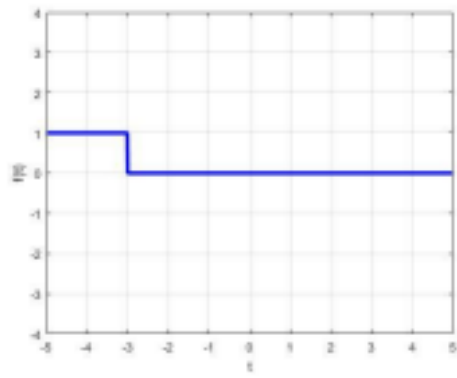
(b)



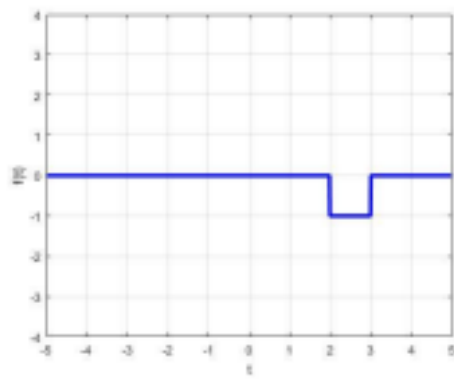
(c)



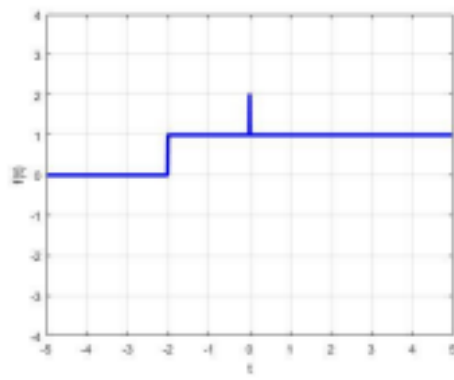
(d)



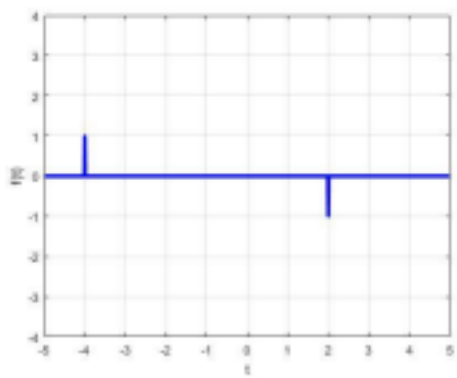
(e)



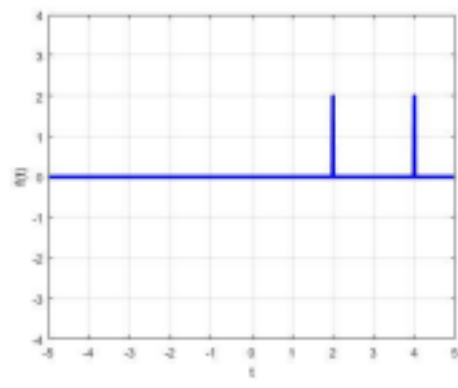
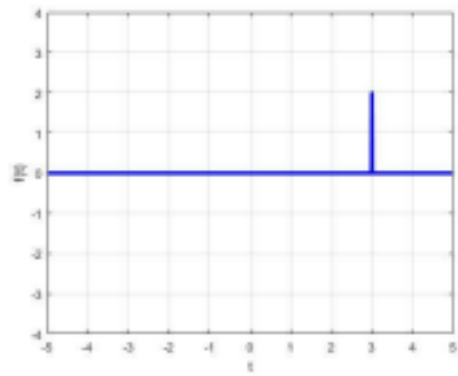
(f)



(g)



(h)



2)

(a) $y(t) = r(t - 2)$

(b) $y(t) = \delta(t - 1) + \delta(t - 2) + \delta(t - 3) + \delta(t - 4) + \delta(t - 5)$

(c) $y(t) = u(t - 1) - u(t - 3) + u(t - 7) - u(t - 9)$

2 Sistemas

1)

(a) É causal

(b) É linear

2)

(a) É linear

(b) É invariante no tempo

3)

(a) Linear e variante no tempo

(b) Não linear e invariante no tempo

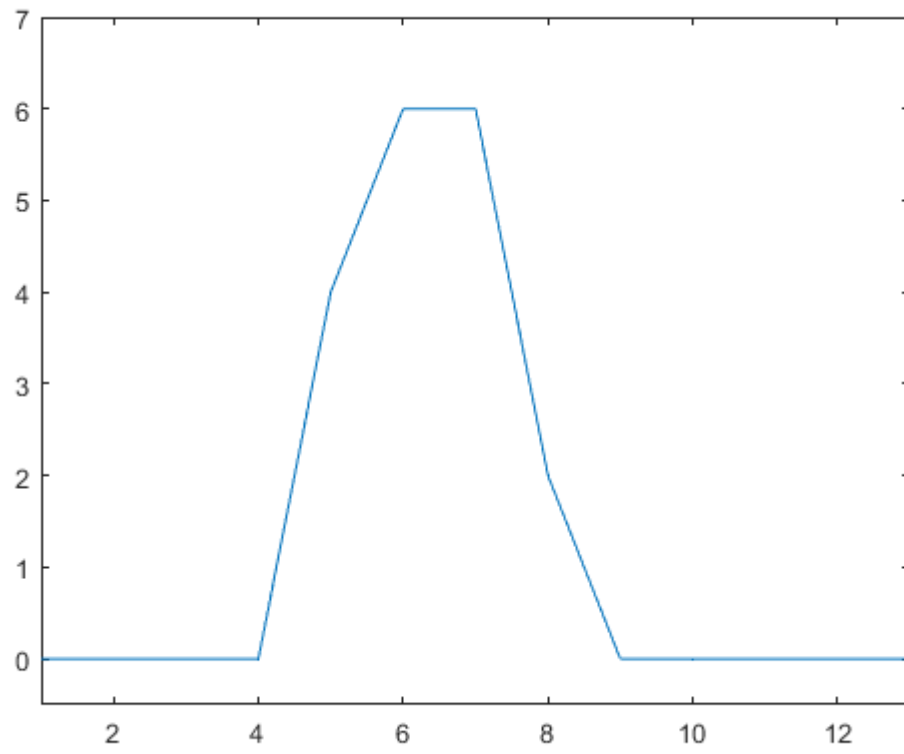
(c) Linear e invariante no tempo

(d) Linear e invariante no tempo

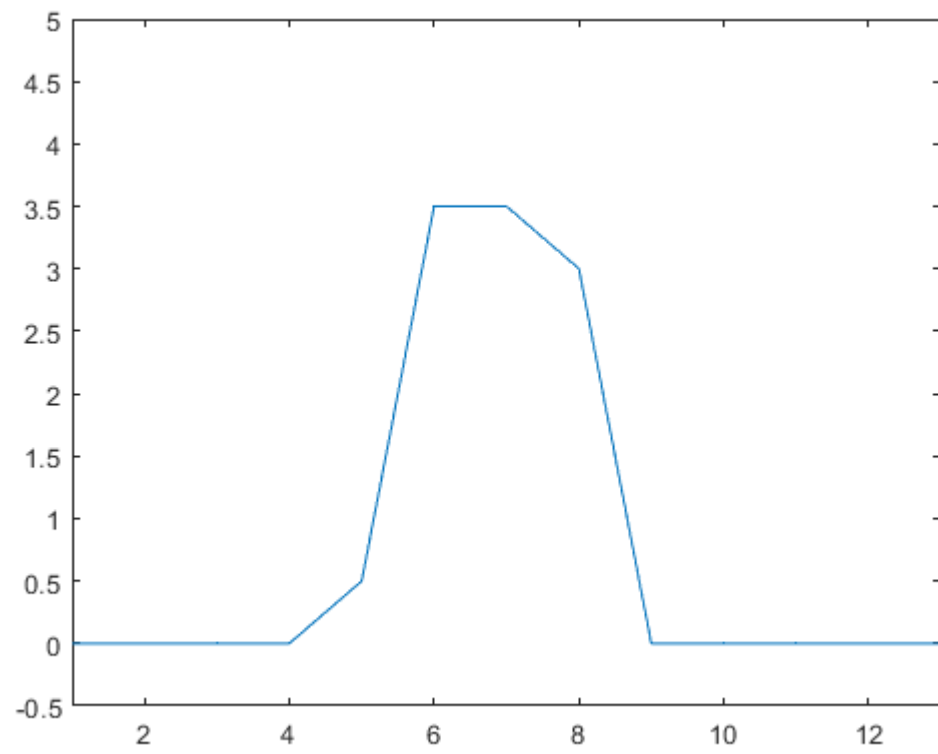
3 Convoluções

1)

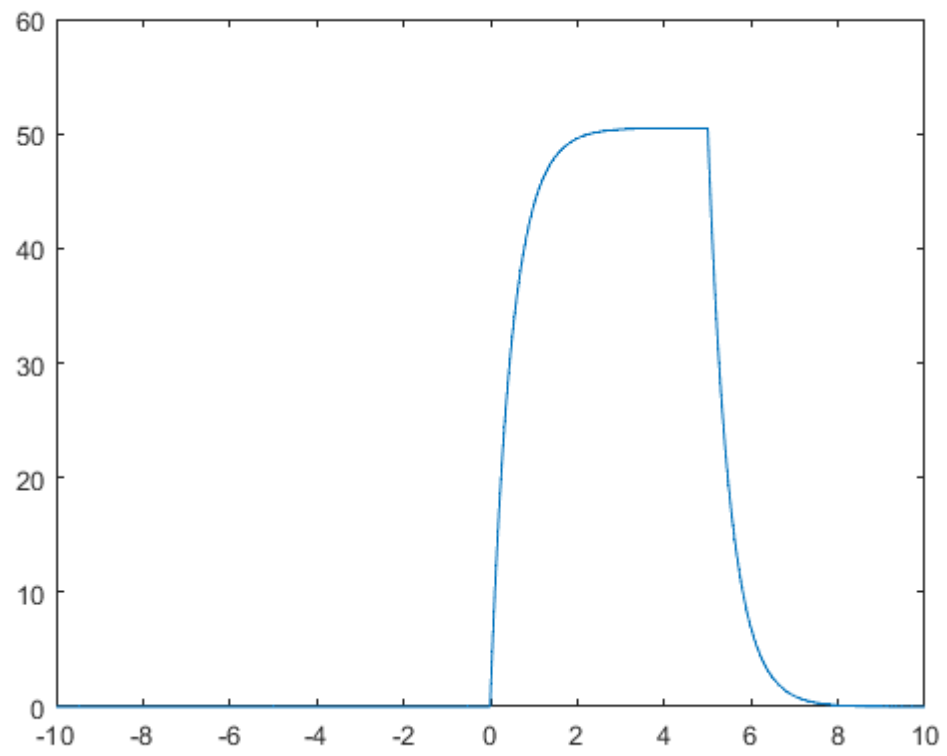
(a)



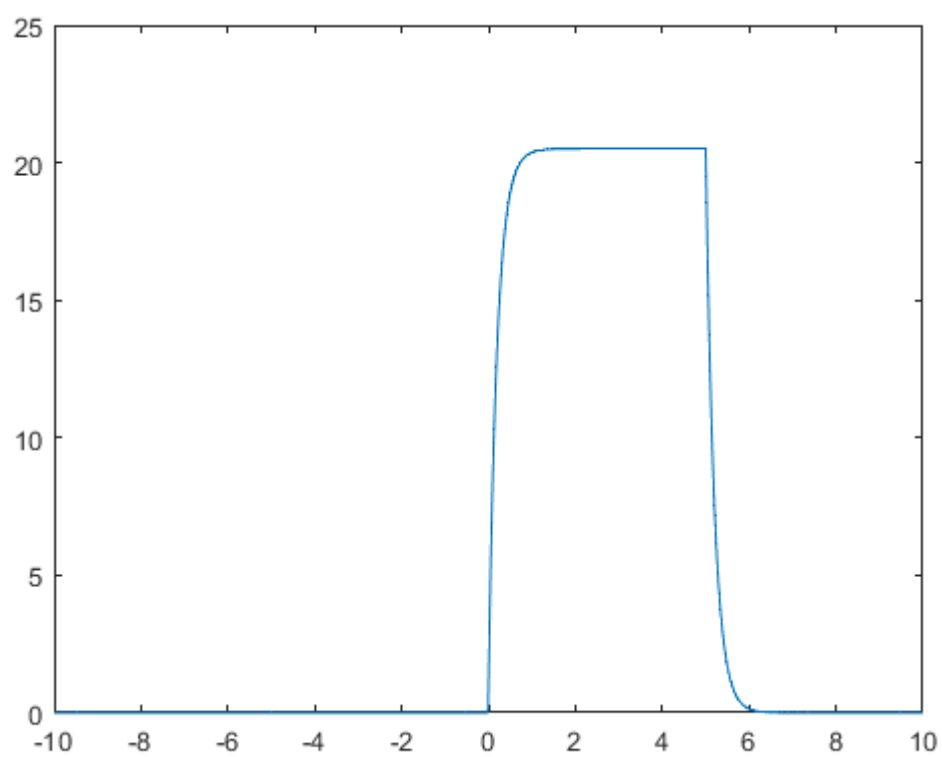
(b)



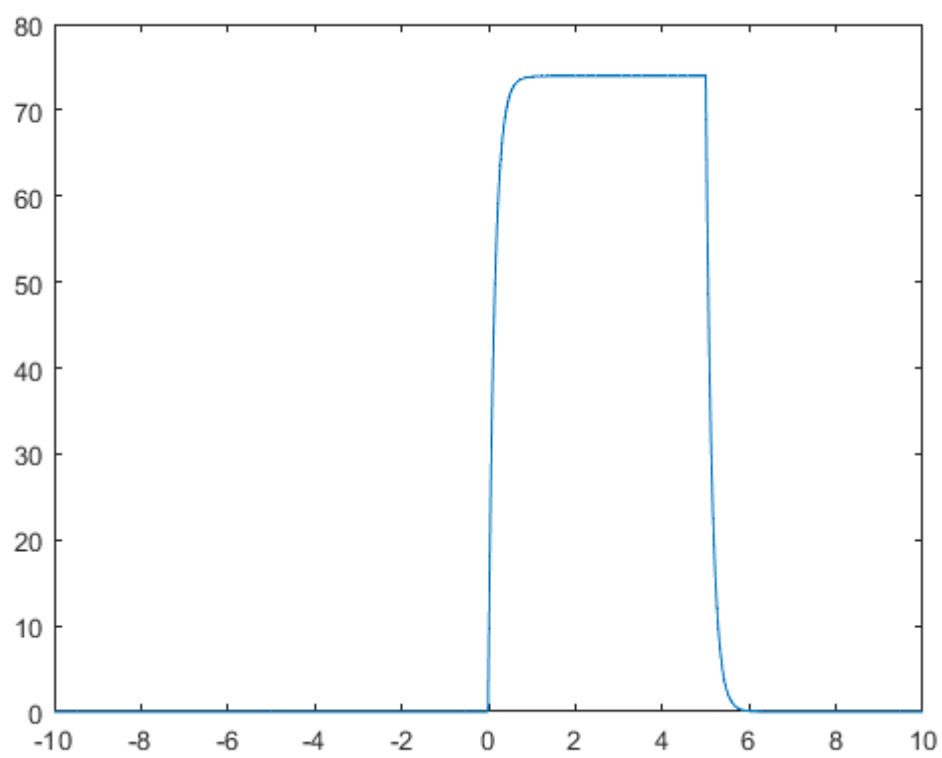
2)
(a)



(b)



(c)



4 Transformada de Fourier

1)

$$(a) F(j\omega) = \frac{e^{-j\omega}}{2+j\omega}$$

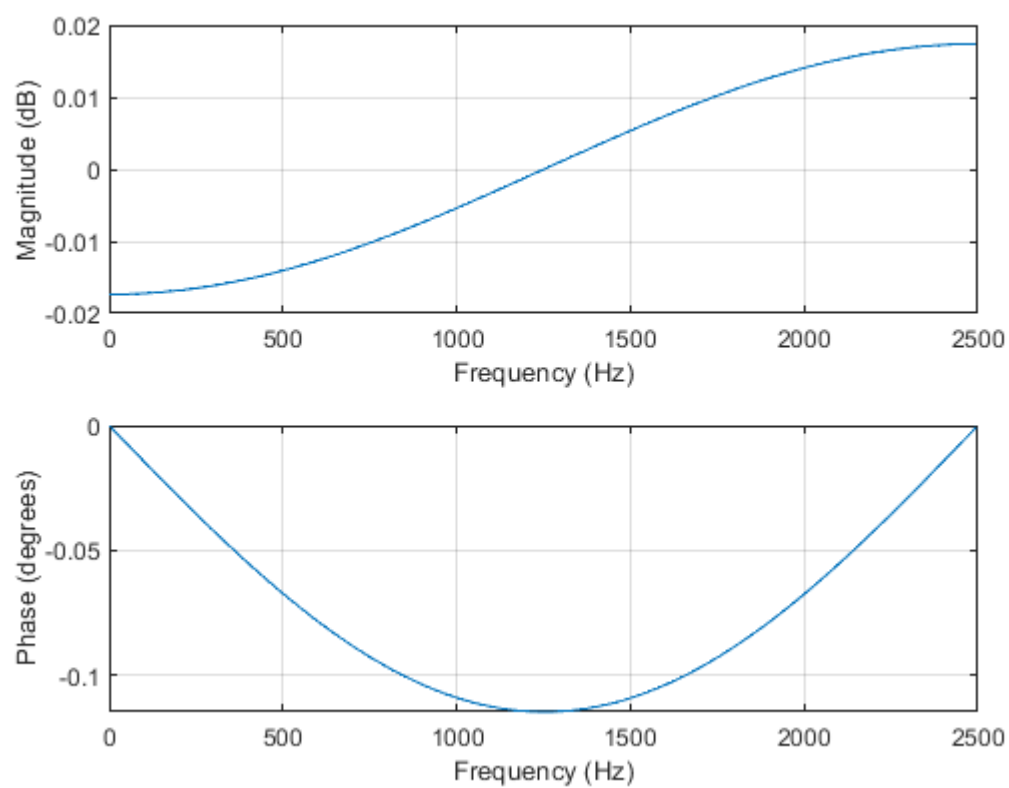
$$(b) F(j\omega) = 2 * \cos(\omega)$$

2)

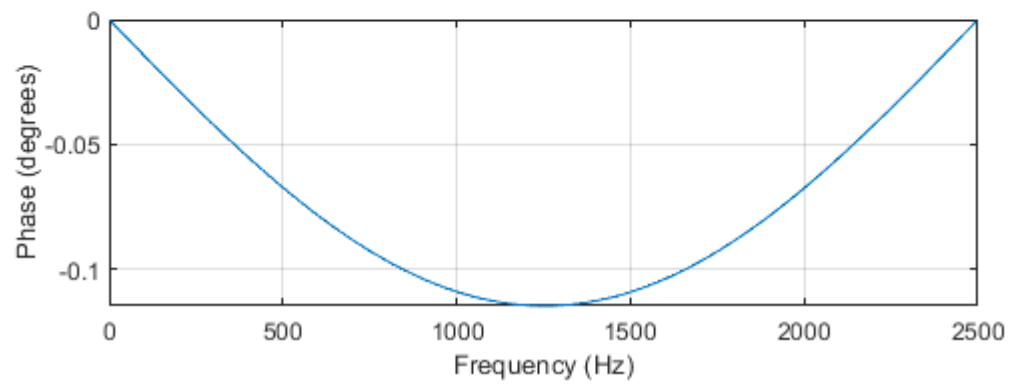
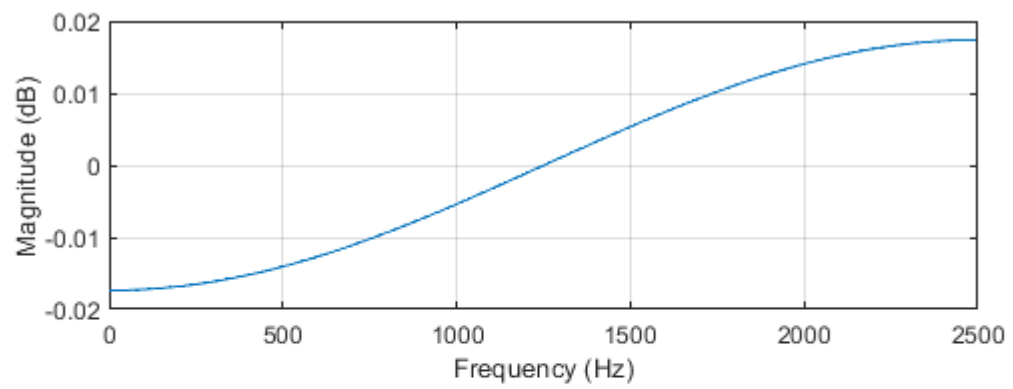
$$(a) f(t) = 1 + \cos(4\pi t)$$

$$(b) f(t) = u(t) - u(t - 1)$$

5 Filtragem Digital



1)



2)

3) Foi utilizado um filtro de ordem 2.

